AI/ML intern DAY 6

TASK 3 — RESEARCH ON CLOTH SEGMENTATION AI

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1. Abstract

This task focused on studying AI-based cloth segmentation systems used in modern fashion technology and virtual try-on applications.

The research explored:

- human parsing systems
- cloth segmentation models
- clothing region extraction
- AI body segmentation
- digital garment analysis

The task also studied how Artificial Intelligence separates clothing regions from the human body using Computer Vision and Deep Learning techniques.

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2. Objective

The objective of this task was to understand how AI-based cloth segmentation systems identify and separate clothing regions from human body images.

The task focused on:

- clothing segmentation workflows
- human parsing systems
- body region extraction
- AI segmentation models
- virtual fitting technologies
- image masking systems

3. Introduction to Cloth Segmentation AI

Cloth segmentation is a Computer Vision process where Artificial Intelligence separates:

- shirts
- pants
- jackets
- dresses
- shoes
- body regions

from an image automatically.

The AI system analyses:

- clothing boundaries
- body contours
- body posture
- garment regions
- image textures

The segmented regions help AI systems generate:

- virtual try-on systems
- fashion analysis systems
- smart shopping platforms
- body parsing applications

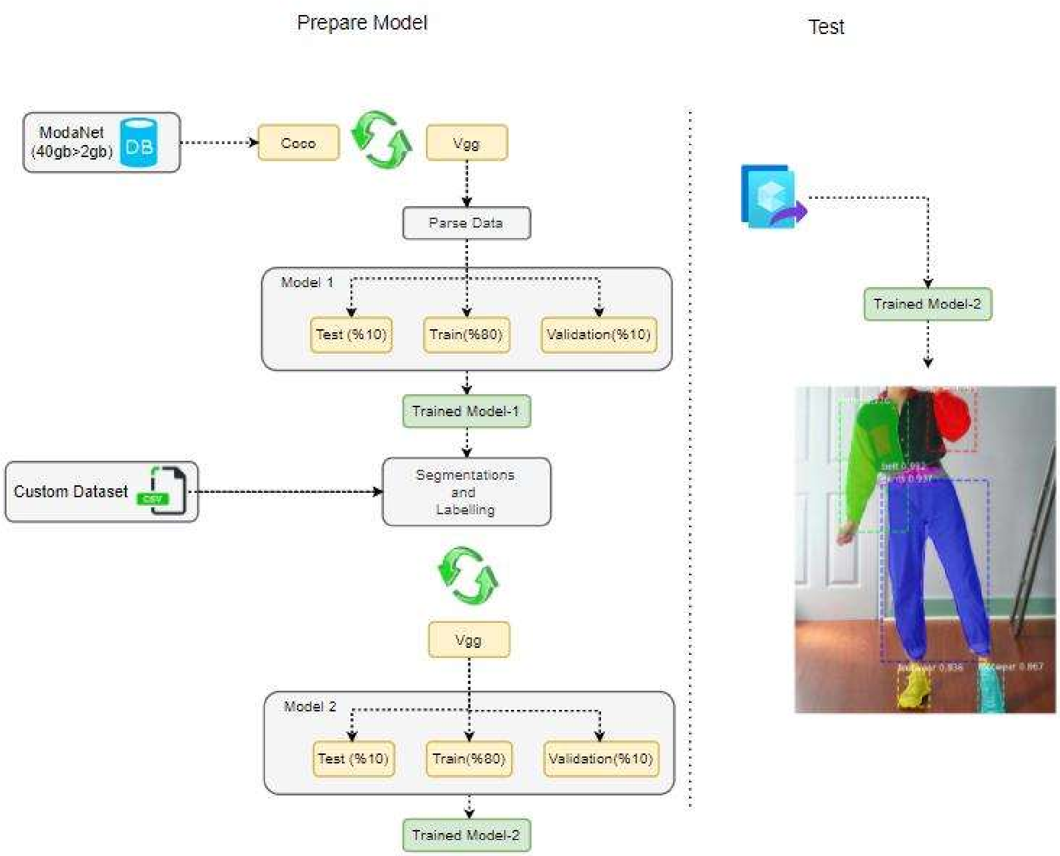


Figure 1 — AI Cloth Segmentation System

4. Workflow of Cloth Segmentation Systems

The cloth segmentation workflow consists of multiple stages.

Stage 1 — Image Input

The AI system receives:

- human body images
- fashion photographs
- clothing samples
- body pose images

Stage 2 — Human Detection

The AI system first detects:

- human body
- body orientation
- body posture
- foreground regions

Stage 3 — Clothing Region Analysis

The segmentation model identifies:

- upper garments
- lower garments
- shoes
- sleeves
- body parts

using Deep Learning and image analysis.

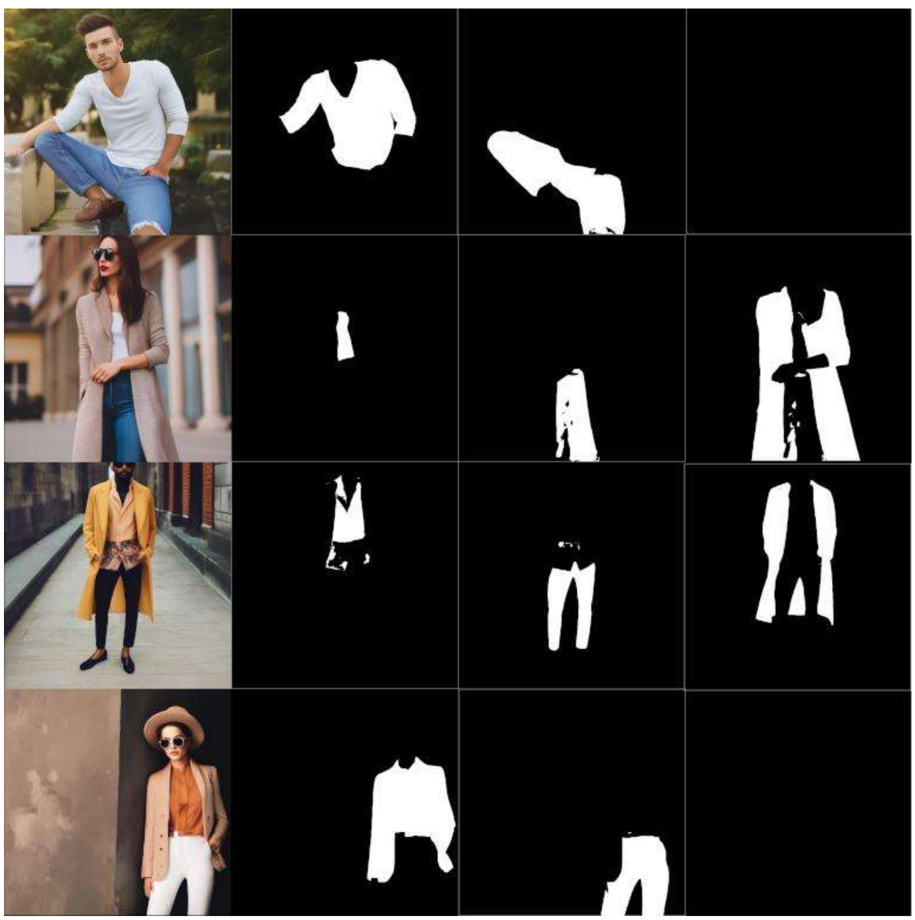
Stage 4 — Segmentation Mask Generation

The AI generates:

- clothing masks
- body masks
- segmentation regions
- body part boundaries

Stage 5 — Final Visualization

The segmented clothing regions are displayed separately from the body image.



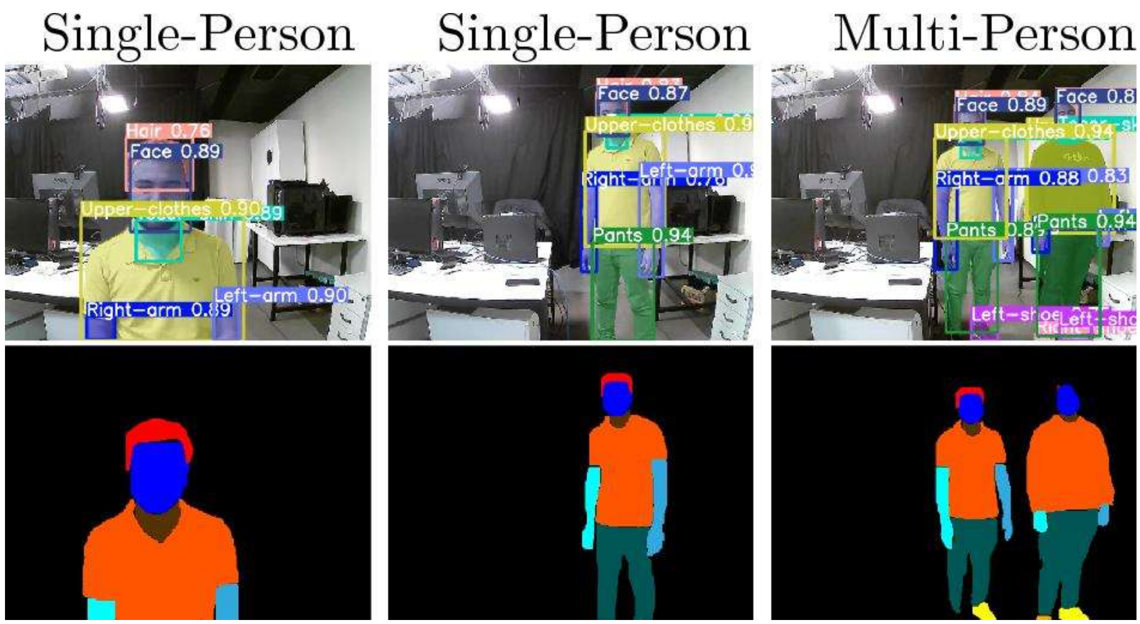


Figure 2 — Clothing Segmentation Output

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5. Technologies Used

Technology	Purpose
Artificial Intelligence	Segmentation analysis
Computer Vision	Image understanding
Deep Learning	Cloth region prediction
OpenCV	Image processing
MediaPipe	Body landmark tracking
Python	AI implementation

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6. Popular AI Segmentation Models

Model	Purpose
MediaPipe Segmentation	Body segmentation
U-Net	Image segmentation
Mask R-CNN	Object segmentation

DeepLabV3	Semantic segmentation
OpenPose	Human body parsing

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7. Pseudo Code — Cloth Segmentation Workflow

START

Load human body image

Detect human body

Identify clothing regions

Generate segmentation masks

Separate body and clothing regions

Render segmented output

Display final segmentation result

END

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8. Algorithm — AI Cloth Segmentation System

Step-by-Step Algorithm

Step 1

Load input image using OpenCV.

Step 2

Detect human body regions.

Step 3

Analyse body posture and body contours.

Step 4

Apply segmentation model.

Step 5

Generate clothing masks.

Step 6

Separate clothing regions from background.

Step 7

Render segmentation output.

Step 8

Display final segmented image.

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9. Improvements in AI Cloth Segmentation

Improvement	Purpose
Better segmentation masks	Improved clothing extraction
Body contour analysis	Accurate body separation
Deep learning models	Better prediction accuracy
Real-time segmentation	Faster visualization
Landmark tracking	Improved body analysis

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10. Applications of Cloth Segmentation AI

Cloth segmentation systems are used in:

- virtual try-on systems
- AI fashion technology
- digital shopping platforms
- body parsing systems
- augmented reality applications
- intelligent fashion systems

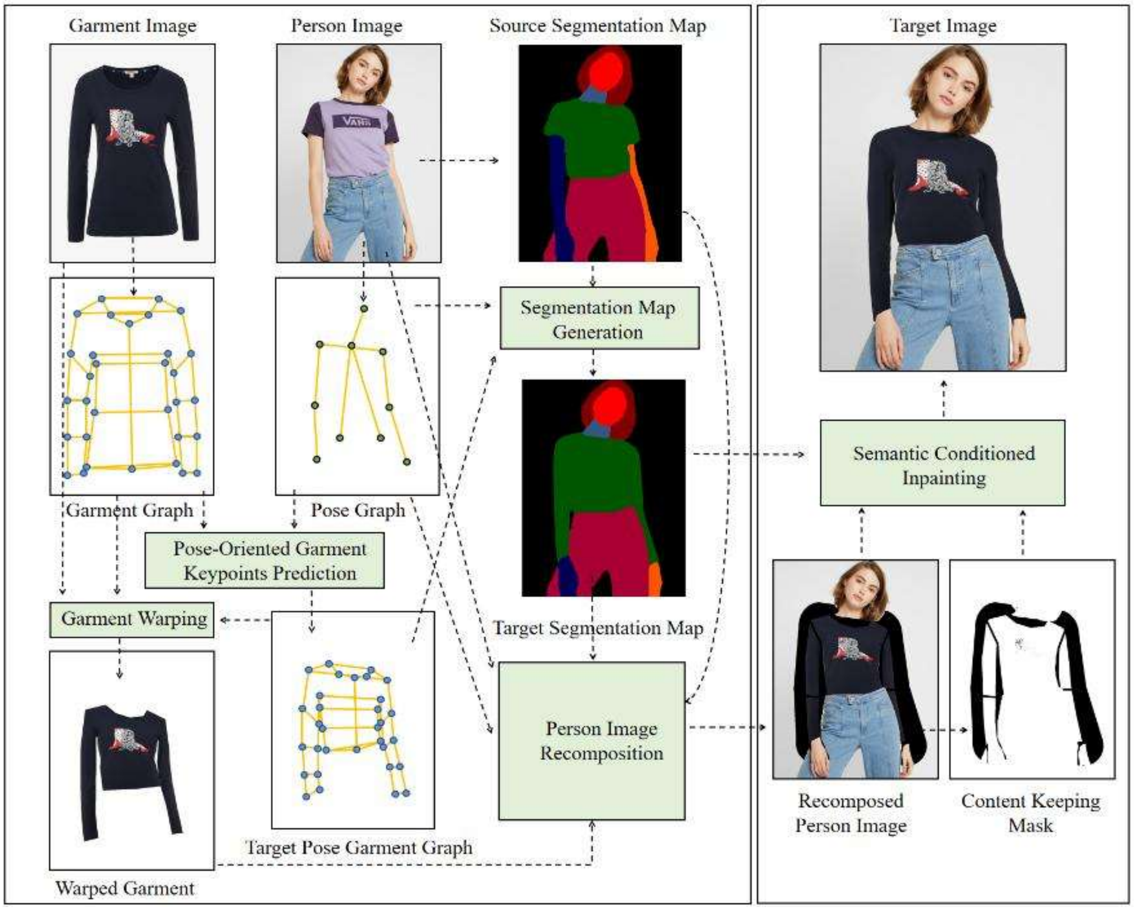

Figure 3 — AI Fashion Segmentation Workflow

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11. Challenges Faced

Challenge	Impact
Complex clothing textures	Reduces segmentation accuracy
Loose clothing	Difficult boundary detection
Similar background colours	Poor segmentation masks
Lighting variations	Affects image quality

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12. Knowledge and Skills Acquired

During this task, the following concepts were understood:

- AI segmentation workflows
- body parsing systems
- cloth region analysis
- image masking techniques
- Deep Learning segmentation
- virtual fitting systems
- computer vision applications

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13. Conclusion

AI cloth segmentation systems play an important role in modern fashion technology and virtual fitting applications using Artificial Intelligence and Computer Vision.

The research improved understanding of:

- cloth segmentation systems
- human parsing workflows
- AI body segmentation
- clothing mask generation
- virtual try-on technologies
- image segmentation models

These systems are widely used in intelligent shopping platforms, AI fashion systems, and digital garment simulation technologies.